THE SECTIONED PICARD STACK IS SMOOTH AND SURJECTIVE OVER THE PICARD STACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition situation-pic (canonical tag 0D25). cited at statement line 3153. Standing situation with a section.
- Section algebraic-section-representable-morphism (canonical tag 04ST). cited at proof line 3167. Framework.
- Section algebraic-section-morphisms-representable-by-algebraic-spaces (canonical tag 04SX). cited at proof line 3168. Framework.
- Section algebraic-section-representable-properties (canonical tag 03YJ). cited at proof line 3169. Framework.
- Lemma algebraic-lemma-criterion-map-representable-spaces-fibred-in-groupoids (canonical tag 02ZY). cited at proof line 3177. Isom-functor criterion.
- Proposition proposition-isom (canonical tag 08K9). cited at proof line 3179. Affine finite presentation of U.

Source reference. Original definition 0D25: sources/stacks-chow/tags.txt, source lines 16370–16370. Cited at statement line 3153: the lemma is phrased in this situation.

Source reference. Original section 04ST: sources/stacks-chow/tags.txt, source lines 6231–6231. Cited at proof line 3167: framework reference.

Source reference. Original section 04SX: sources/stacks-chow/tags.txt, source lines 6235–6235. Cited at proof line 3168: framework reference.

Source reference. Original section 03YJ: sources/stacks-chow/tags.txt, source lines 5186–5186. Cited at proof line 3169: framework reference.

Source reference. Original lemma 02ZY: sources/stacks-chow/tags.txt, source lines 3929–3929. Cited at proof line 3177: the isomorphism-functor description.

Source reference. Original proposition 08K9: sources/stacks-chow/tags.txt, source lines 10877–10877. Cited at proof line 3179: representability by an affine finitely presented U.

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Lemma 0.2. *In Situation 0D25 let $\sigma : B \to X$ be a section. The morphism $\mathcal{P}ic_{X/B,\sigma} \to \mathcal{P}ic_{X/B}$ is representable, surjective, and smooth.*

Proof. Let T be a scheme and let $(Sch/T)_{fppf} \to \mathcal{P}ic_{X/B}$ be given by the object $\xi = (T, h, \mathcal{L})$ of $\mathcal{P}ic_{X/B}$ over T. We have to show that

$$(Sch/T)_{fppf} \times_{\xi, \mathcal{P}ic_{X/B}} \mathcal{P}ic_{X/B,\sigma}$$

is representable by a scheme V and that the corresponding morphism $V \to T$ is surjective and smooth. See Algebraic Stacks, Sections 04ST, 04SX, and 03YJ. The forgetful functor $\mathcal{P}ic_{X/B,\sigma} \to \mathcal{P}ic_{X/B}$ is faithful on fibre categories and for T'/T the set of isomorphism classes is the set of isomorphisms

$$\alpha' : \mathcal{O}_{T'} \longrightarrow (T' \to T)^* \sigma_T^* \mathcal{L}$$

See Algebraic Stacks, Lemma 02ZY. We know this functor is representable by an affine scheme U of finite presentation over T by Proposition 08K9 (applied to id : $T \to T$ and $\mathcal{O}_T$ and $\sigma^*\mathcal{L}$). Working Zariski locally on T we may assume that $\sigma_T^*\mathcal{L}$ is isomorphic to $\mathcal{O}_T$ and then we see that our functor is representable by $\mathbf{G}_m \times T$ over T. Hence $U \to T$ Zariski locally on T looks like the projection $\mathbf{G}_m \times T \to T$ which is indeed smooth and surjective. □